AI IN BUSINESS

# Smart Factory Automation with AI & Robotics

mobica Automotive Glass Manufacturing Factory — Egypt


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|---|---|
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# Presentation Agenda


**01 Industry & Business Context**
Market landscape, factory profile & competitive environment

**02 Problem Statement**
Six operational gaps and why traditional methods failed

**03 Data Analytics**
Data types, KPIs, and six data mining techniques

**04 AI Solution Overview**
Computer vision, predictive maintenance, ML models & RPA

**05 Analytical Techniques**
Descriptive, diagnostic, predictive & prescriptive analytics

**06 Digital Transformation**
Process re-engineering, automation & cultural change

**07 ERP Integration**
SAP S/4HANA modules, data flow, BPM & flowcharts

**08 Business Impact & ROI**
$9.5M net annual benefit · 193% ROI · 5.3-month payback

**09 Risks, Ethics & ISO 27001**
Data security, AI bias, ethical commitments & limitations

**10 Recommendations & Future Vision**
Near-term investments and 2033 strategic ambition

# Global Automotive Glass Market

**$32B**

Global Market
Annual Value

**ECE R43**

Mandatory Safety
Certification

**MENA**

Key Growth Region
for Localisation

### Key Players

Saint-Gobain Sekurit · AGC Inc. · Fuyao Glass · Guardian Industries collectively dominate global output

### Industry 4.0 Drive

Automation & data-driven ops are now an expectation, not an advantage. Late movers risk OEM exclusion

### MENA Opportunity

Egypt & Morocco localisation drives create real openings — but only for manufacturers who can compete on quality

# mobica Factory Profile

| | |
|---|---|
| **Location** | Industrial Zone, 6th of October, Egypt |
| **Established** | 2025 — Greenfield Investment |
| **Annual Capacity** | 700,000 units/year (100,000 full vehicle sets) |
| **Products** | Laminated windshields · Tempered side/rear glass · Sunroof panels |
| **Total Investment** | $20 million (including $4.2M AI/automation) |
| **Workforce** | 90 direct workers + 30 management & technical staff |
| **Certifications** | ISO 9001 · IATF 16949 · E-Mark ECE R43 · ISO 27001:2022 |
| **Market Split** | Egypt 40% · Turkey 35% · Morocco 25% |
| **Key Customers** | Nissan Egypt · GME · Ghabbour Auto · MG · EIT · Proton · Aftermarket |

# Six Operational Challenges Before AI

mobica was running a 20th-century factory in a 21st-century market

### 12.5% Defect Rate

6 manual inspectors/shift; fatigue-driven errors; 45 complaints/month; excluded from OEM contracts

### 18% Unplanned Downtime

No predictive systems; 24–72hr halts per failure; $3.2M production loss annually

### 17.5% Material Waste

Experience-based cutting layouts with no optimisation algorithm; $2.1M annual raw material loss

### 8–12 Injuries per Year

45 workers manually handling 80–120 kg glass sheets with zero automation

### No Real-Time Visibility

Paper records & Excel only; management had a 1-week blind spot on all operational KPIs

### $45.20/Unit Cost

Above both Chinese ($28–32) and European ($38–42) rivals — no competitive position

# Three Gaps That Had to Close

| 01 Operational Gap | 02 Strategic Gap | 03 Financial Gap |
|---|---|---|
| **OEE 68%** | **$3–5M** | **$45.20/unit** |
| vs. industry benchmark 85%+<br>~180,000 units of foregone annual output | OEM contracts fully inaccessible<br>due to quality & delivery failures | ~20% above competitive threshold<br>no price advantage vs. China or Europe |

*These three gaps reinforced each other in a self-sustaining cycle — only a solution addressing all three simultaneously could break the pattern.*

# Data Sources & KPI Dashboard

**60%**

### Structured Data

SAP records: 1M+ work orders, 500K QM entries, IoT sensor feeds

**35%**

### Unstructured Data

500K defect photos + maintenance notes & shift handover reports

**5%**

### Real-Time Streams

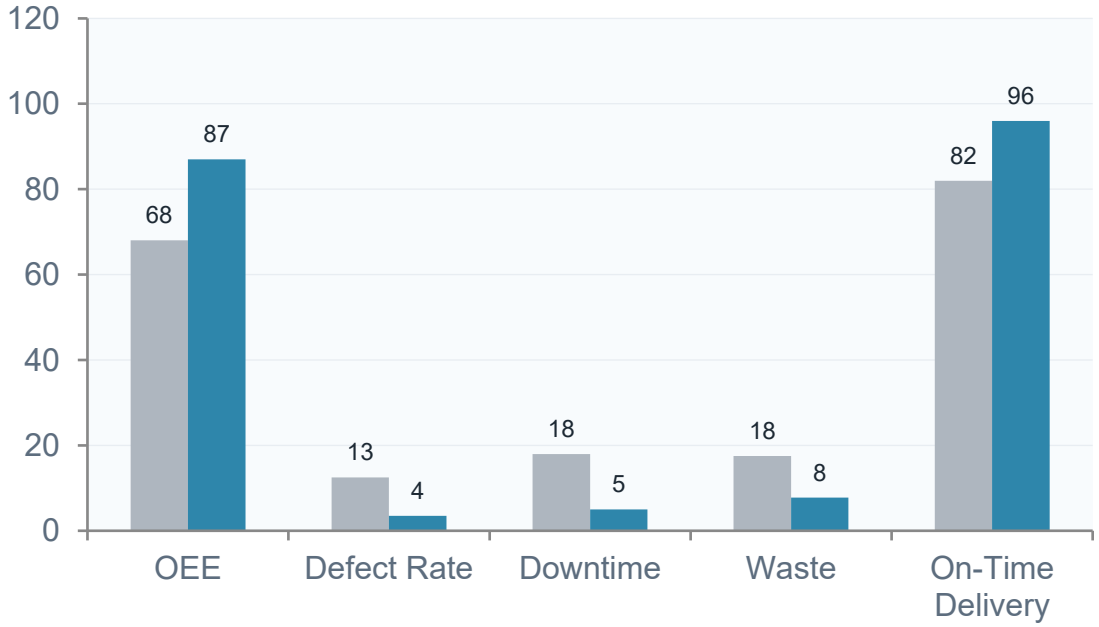
200GB/day sensor data · 30fps cameras · production counters

*Unified via Apache Kafka (100K msg/s) → 150TB Azure Data Lake → Power BI live dashboards*

### KPI Dashboard — Before vs. After AI

| KPI | Before AI | After AI |
|---|---|---|
| OEE | 68% | 87% |
| Defect Rate | 12.5% | 3.5% |
| Unplanned Downtime | 18% | 5% |
| Material Waste | 17.5% | 7.8% |
| Cost per Unit | $45.20 | $32.80 |
| On-Time Delivery | 82% | 96% |
| Cust. Satisfaction | 3.8/5 | 4.6/5 |

# KPI Improvement — Before vs. After AI




**−72%**
Defect Rate

**+28%**
OEE

**−55%**
Waste

**−27%**
Cost/Unit

**+17%**
On-Time Del.

**96%**
OTD Target

# Decision-Making via Data Mining

6 techniques driving every key decision in the factory

### Association Rules

Apriori — 89% confidence link between summer humidity & lamination defects → dehumidifier installation.

### Clustering

K-Means (k=7) & DBSCAN — revealed Monday bubble-defect pattern linked to uneven weekend glass cooling.

### Time Series Forecasting

ARIMA + Prophet + LSTM — demand accuracy: 65% → 92%; drives procurement & workforce planning.

### Anomaly Detection

Isolation Forest & Autoencoder — flags sensor deviations 24–48hrs before conventional alarms; stops breakdowns pre-emptively.

# Four Core AI Systems

$4.2M programme — 18 months — January 2025 to June 2026

## Computer Vision — Quality

- ResNet-50 CNN · 100K labelled defect images · 15 categories
- 4 × 8MP cameras · pass/fail in <3 sec · 99.5% accuracy
- NVIDIA Jetson edge hardware · defect rate: 12.5% → 3.5%

## Predictive Maintenance

- Random Forest + LSTM ensemble (60%/40% weighted vote)
- 24-hour rolling sensor window · 48–72hr advance warning
- MTBF: 480 → 850 hrs (+77%) · downtime: 18% → 5%

## Cutting Optimisation & RPA

- Genetic Algorithm (500 generations) + RL scheduling agent
- 7 robots + 5 AGVs · Siemens PLC S7-1500 coordination
- Waste: 17.5% → 7.8% · 45 handlers → 8 operators (reskilled)

## Demand Forecasting

- Facebook Prophet + LSTM hybrid · 8% MAPE
- Weekly batch refreshed daily · 3 export markets
- Forecast accuracy: 65% → 92%

# ML Taxonomy & SDLC Methodology

## Machine Learning Approach

### Supervised

Quality CNN (ResNet-50, 100K imgs, 70/15/15 split, ImageNet TL) + Predictive Maintenance Random Forest on labelled sensor-health pairs

### Unsupervised

Anomaly Detection Autoencoder (128→64→32→16 nodes) + K-Means & DBSCAN surfacing Monday bubble pattern & other root causes

### Semi-Supervised

Cutting optimisation: 1K expert layouts → pseudo-labels for 100K cuts (≥95% confidence) → Genetic Algorithm (500 gen) → −55% waste

## SDLC: Agile-Waterfall Hybrid (18 months)

| | | |
|---|---|---|
| **1 – Planning** | **Mo 1–2** | Budget & charter approved |
| **2 – Analysis** | **Mo 3–4** | Requirements & use cases |
| **3 – Design** | **Mo 5–6** | Architecture & SAP integration |
| **4 – Development** | **Mo 7–12** | 16 Agile sprints · CI/CD |
| **5 – Testing** | **Mo 13–15** | SIT, UAT, ISO 27001 audit |
| **6 – Deployment** | **Mo 16–18** | Staged go-live · 40hr training |
| **7 – Maintenance** | **Ongoing** | Quarterly retraining · 99.5% |

# Four-Category Analytics Framework

From describing what happened to prescribing what to do

| | | | |
|---|---|---|---|
| 01 | **Descriptive** *What happened?* | Tools: Power BI · SAP Analytics · pandas | Real-time OEE dashboards · defect distribution · shift summaries |
| 02 | **Diagnostic** *Why did it happen?* | Tools: scikit-learn · Minitab · R | Monday bubble pattern · Furnace B degradation root-cause identified |
| 03 | **Predictive** *What will happen?* | Tools: TensorFlow · Azure ML · LSTM · Prophet | Equipment failure 48–72hr ahead · demand 92% accuracy · quality risk flagged |
| 04 | **Prescriptive** *What should we do?* | Tools: OR-Tools · CPLEX · AnyLogic | Cutting layout −55% waste · scheduling +25% · AGV route optimisation |

# Process Re-Engineering & Cultural Change

## 5 Re-Engineered Core Processes

| | | |
|---|---|---|
| **Production Planning** | Weekly Excel → AI scheduler + SAP PP MRP | **Order cycle: 12–15 days → 5–7 days** |
| **Quality Control** | 6 manual inspectors → automated CV station | **Defect rate: 12.5% → 3.5%** |
| **Maintenance** | Reactive → predictive (48–72hr window) | **Downtime: 18% → 5%** |
| **Inventory Mgmt.** | Weekly manual count → AGV/RFID real-time | **Stockouts: 8% → 0.8%** |
| **Order-to-Delivery** | Estimate-based → live ATP engine | **On-time delivery: 82% → 96%** |

## 4 Cultural Shifts

**Instinct → Evidence**

Data anchors all decisions

**Reaction → Anticipation**

48–72hr predictions replace crisis management

**Silos → Shared Visibility**

One Power BI dashboard for all levels

**Centralised → Distributed**

Operators self-adjust with real-time data

# SAP S/4HANA — 6 Integrated Modules

## SAP PP

### Production Planning

AI forecasts feed MRP; AI scheduler proposes work order sequences; IoT confirms completions

✓ **Planning: weekly → real-time**

## SAP QM

### Quality Management

Vision system defects auto-raise QM notifications with images & root-cause data

✓ **Reporting lag: 1 week → instant**

## SAP MM

### Materials Management

AGV/RFID post goods transactions automatically; AI auto-raises purchase requisitions

✓ **Stockouts: 8% → 0.8%**

## SAP PM

### Plant Maintenance

Predictive model raises PM work orders with failure mode, spare parts & optimal window

✓ **Maintenance cost ↓35%**

## SAP SD

### Sales & Distribution

Real-time ATP check per order; delivery dates committed algorithmically; 98% docs auto-generated

✓ **On-time delivery: 82% → 96%**

## SAP FI/CO

### Finance & Controlling

Every production event posts to accounts in real time; profitability by product/customer live

✓ **Month-close: 10 days → 3 days**

# BPM Swimlane — End-to-End Production Process

| | |
|---|---|
| **Customer / SAP SD** | [START] Order Received → ATP Check (Live Capacity) → Delivery Date Confirmed → Sales Order in SAP SD → Released to Production |
| **Production Planning (PP + AI)** | AI Demand Forecast → MRP Run → AI Scheduler Optimises Sequence → Work Orders Generated → Material Requisition (SAP MM) |
| **Manufacturing (Robots + IoT)** | ABB Robot Loads → KUKA CNC Cutting → Tempering 700°C → IoT Streams → [Maintenance Alert?] YES→PM Work Order / NO→Continue → Confirm in SAP PP |
| **Quality Control (CV-AI + QM)** | 4-Camera Array → CNN ResNet-50 (<3 sec) → [Pass?] FAIL→QM Notification+Reject / PASS→Certificate of Analysis Auto-Generated → Batch Released |
| **Warehouse / Logistics (AGV)** | AGV Transport → RFID Posts Goods Receipt (SAP MM) → Stock Updated Real-Time → Pick & Pack → Export Docs Auto-Generated → Shipment → [END] |

# Financial Impact & Return on Investment

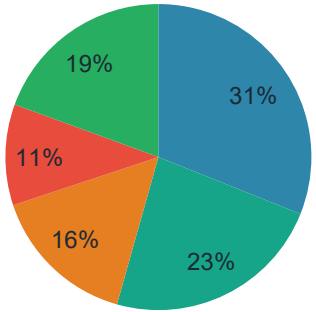
**$4.2M**
Total Investment

**$9.5M**
Net Annual Benefit

**5.3 mo**
Payback Period

**193%**
Annual ROI

$10.3M Annual Cost Savings




**EBITDA & Key Financials**

| | | |
|---|---|---|
| Before AI: | $2.2M | 12% margin |
| After AI: | $11.7M | 34% margin |
| 3-Yr NPV: | $19.4M | @10% WACC |
| 3-Yr ROI: | 579% | 193%/yr avg |

# ISO 27001:2022, AI Bias & Ethical Commitments

## ISO 27001:2022 — 4 Relevant Clauses

### A.8.2 – Information Classification

AI model weights = Confidential; production records = Internal; marketing = Public

### A.12.4 – Logging & Monitoring

Splunk SIEM; 90-day log retention; automated alerts on anomalous access patterns

### A.12.6 – Vulnerability Management

Monthly scans; critical patches ≤48hrs; annual independent penetration test

### A.13.1 – Network Security

Purdue Model OT/IT physical segmentation; VPN + MFA for all remote access

## Ethics, Bias & Limitations

### AI Model Bias

Bubble defects = 28% of training data; monthly per-category audits + mandatory human review at 95–98% confidence

### Workforce Ethics

30 positions automated → no redundancy; 6-month funded reskilling; +15% average wages; 5 new specialist roles

### Algorithmic Transparency

XAI surfaces defect image regions; full decision audit trail; unconditional human override on all AI outputs

### Current Limitations

$4.2M threshold excludes SMEs; 11% maintenance prediction error; Azure/SAP vendor lock-in risk

# Near-Term Investments & 2033 Vision

### Energy AI

Off-peak tariff scheduling for furnaces
Saving: $0.6M/year

### 3D Vision System

ADAS/EV glass curvature measurement
Investment: ~$0.4M

### Digital Twin

Risk-free scenario testing
30% faster innovation cycles (~$0.8M)

### Blockchain Tracing

Tamper-proof genealogy for Tier-1 OEM
Investment: ~$0.2M

## Strategic Roadmap — 2033

**Phase 2**
**2027–28**

Morocco facility
AMR robots + 5G edge

**Phase 3**
**2029+**

Lights-out production
Quantum optimisation

**2033 Target**

3 factories · 2.1M units/yr
25% MENA share · 35%+ EBITDA